

EXAMPLE

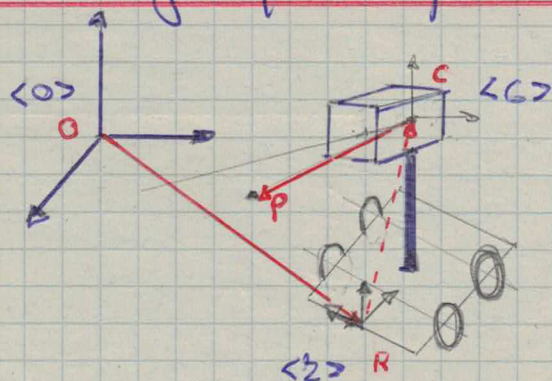
A mobile robot is supporting an RGBD camera -
The camera reference frame is calibrated w.r.t. the robot's ref frame $\langle r \rangle$ -

The camera is assumed fixed w.r.t. robot.

The robot is moving w.r.t. a frame $\langle o \rangle$ and its attitude (orientation), position, linear/angular velocities and accelerations are assumed to be known (*) at some time t_m^n

A moving point P is measured by the camera (**) whose output at any measurement time t_m^n is the position of P w.r.t. $\langle c \rangle$ - Along with the position of P also velocity and acceleration of P w.r.t. $\langle c \rangle$ are assumed to be known (*) at time t_m^n

We are interested in predicting position/velocity/acceleration of P w.r.t. $\langle o \rangle$ for some time interval s.t. $t > t_m^n$

Modeling of the problem

Sketch of the scenario

(**) Image processing methods to extract P from the image stream is not part of the discussion here -

(*) Either measured, estimated or computed

Let $P(t)$ the movement of P in space -
 We are interested in computing

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$$(P(t) - 0) = \underline{z}_{P/0}$$

$$\frac{d}{dt} \underline{z}_{P/0} = \underline{v}_{P/0}$$

$$\frac{d}{dt} \underline{v}_{P/0} = \underline{a}_{P/0}$$

given $\underline{z}_{P/0}(t_m), \underline{v}_{P/0}(t_m), \underline{a}_{P/0}(t_m)$ -

If the motion of $P(t)$ is "smooth" (*) then

$$\underline{z}_{P/0}(t) \cong \underline{z}_{P/0}(t_m) + \underline{v}_{P/0}(t_m)(t-t_m) + \frac{1}{2} \underline{a}_{P/0}(t_m)(t-t_m)^2$$

Taylor
expansion (**)

In order to compute numerically the terms the
 eq. above has to be projected on $\langle 0 \rangle$ (as shown later)

$$\underline{z}_{P/0} = \overbrace{\underline{z}_{P/C} + \underline{z}_{C/R}}^{\underline{z}_{P/R}} + \underline{z}_{R/0}$$

↑
measured
by the camera

↑
constant known by
camera calibration

↑
computed/measured
(e.g. by robot "controller")

(**) Used as a possible example of "extrapolation"

(*) A smooth function is intended as any function
 with continuous derivatives up to some order (typic. 2)

$$\underline{v}_{p/o} = \frac{d_o}{dt} [\underline{z}_{p/r} + \underline{z}_{r/o}] =$$

$$= \underbrace{\frac{d_r}{dt} \underline{z}_{p/r}}_{\underline{v}_{p/r}} + \underline{w}_{r/o} \times \underline{z}_{p/r} + \underbrace{\frac{d_o}{dt} \underline{z}_{r/o}}_{\underline{v}_{r/o}}$$

computed/measured
(e.g. by robot controller)

$$\underline{v}_{p/r} = \frac{d_r}{dt} [\underline{z}_{p/c} + \underline{z}_{c/r}]$$

$$= \underbrace{\frac{d_c}{dt} \underline{z}_{p/c}}_{\underline{v}_{p/c}} + \underline{w}_{c/r} \times \underline{z}_{p/c} + \underbrace{\frac{d_r}{dt} \underline{z}_{c/r}}_{\underline{v}_{c/r}}$$

these are 0 if
the camera is fixed
on the robot

Then (assuming a camera fixed*)

$$\underline{v}_{p/o} = \underline{v}_{p/c} + \underline{w}_{r/o} \times [\underline{z}_{p/o} + \underline{z}_{c/r}] + \underline{v}_{r/o}$$

$$\underline{a}_{p/o} = \frac{d_o}{dt} \underline{v}_{p/c} + \frac{d_o}{dt} \left\{ \underline{w}_{r/o} \times [\underline{z}_{p/o} + \underline{z}_{c/r}] \right\} + \underbrace{\frac{d_o}{dt} \underline{v}_{r/o}}_{\substack{\text{computed/measured} \\ \underline{a}_{r/o}}} =$$

(*) Complete the example assuming a moving camera [homework]

$$= \underbrace{\frac{d_c}{dt} \underline{U}_{p/c}}_{\substack{\uparrow \\ \text{measured/computed} \\ \text{at image processing level}}} + \underbrace{\underline{W}_{c/o}}_{\substack{\uparrow \\ \underline{W}_{c/r} + \underline{W}_{r/o} \\ \text{for a fixed camera}}} \times \underline{U}_{p/c} +$$

$$+ \underbrace{\left(\frac{d_o}{dt} \underline{W}_{r/o} \right)}_{\substack{\uparrow \\ \text{for a fixed camera}}} \times \left[\underline{z}_{p/c} + \underline{z}_{c/r} \right] + \underline{W}_{r/o} \times \frac{d_o}{dt} \left[\underline{z}_{p/c} + \underline{z}_{c/r} \right] +$$

$$+ \underline{a}_{r/o}$$

$$\begin{aligned} & \frac{d_c}{dt} \underline{z}_{p/c} + \underline{W}_{c/o} \times \underline{z}_{p/c} + \\ & \frac{d_r}{dt} \underline{z}_{c/r} + \underline{W}_{r/o} \times \underline{z}_{c/r} \\ & \text{for fixed camera} \end{aligned}$$

Then :

$$\underline{z}_{p/o} = \underline{z}_{p/c} + \underline{z}_{c/r} + \underline{z}_{r/o}$$

$$\underline{U}_{p/o} = \underline{U}_{p/c} + \underline{W}_{r/o} \times \left[\underline{z}_{p/c} + \underline{z}_{c/r} \right] + \underline{U}_{r/o}$$

$$\begin{aligned} \underline{a}_{p/o} = \underline{a}_{p/c} + 2 \underline{W}_{r/o} \times \underline{U}_{p/c} + \frac{d_o}{dt} \underline{W}_{r/o} \times \left[\underline{z}_{p/c} + \underline{z}_{c/r} \right] + \\ + \underline{W}_{r/o} \times \left[\underline{W}_{c/o} \times \left(\underline{z}_{p/c} + \underline{z}_{c/r} \right) \right] \end{aligned}$$

This completes the modelling (

Computations (*) (**)

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$${}^0\mathbf{z}_{p/o} = {}^0\mathbf{R} \left[{}^c\mathbf{R} {}^c\mathbf{z}_{p/c} + {}^R\mathbf{z}_{c/r} \right] + {}^0\mathbf{z}_{R/o}$$

measured/
computed (e.g.
inteq. of
Poisson Formula)

constant
(calibration)

measured
(sw. recog. and
image proc.)

constant
(calibration)

measured/computed
(it could be acquired
in robot coordinates
then it must projected
into <0>)

$${}^0\mathbf{R} \cdot {}^c\mathbf{R}$$

$${}^0\mathbf{v}_{p/o} = {}^c\mathbf{R} {}^c\mathbf{v}_{p/c} + [{}^0\mathbf{W}_{R/o} \times] \left[{}^c\mathbf{R} {}^c\mathbf{z}_{p/c} + {}^R\mathbf{z}_{c/r} \right] + {}^0\mathbf{v}_{R/o}$$

measured
(sw. image acq.
and processing)

measured/computed
(it could be known
as ${}^R\mathbf{v}_{R/o}$)

$$\begin{aligned} {}^0\mathbf{a}_{p/o} = & {}^c\mathbf{R} {}^c\mathbf{a}_{p/c} + 2[{}^0\mathbf{W}_{R/o} \times] {}^c\mathbf{R} {}^c\mathbf{v}_{p/c} + \left[\frac{d}{dt} {}^0\mathbf{W}_{R/o} \times \right] \left[{}^c\mathbf{R} {}^c\mathbf{z}_{p/c} + {}^R\mathbf{z}_{c/r} \right] \\ & + [{}^0\mathbf{W}_{R/o} \times] [{}^0\mathbf{W}_{R/o}] \left[{}^c\mathbf{R} {}^c\mathbf{z}_{p/c} + {}^R\mathbf{z}_{c/r} \right] \end{aligned}$$

(**) The camera is assumed fixed on the robot.

(*) Computations are assumed to be performed at sampling (or measurement time). Computation time and accuracy are neglected here. Sampling time is usually much smaller than sampling period.

NOTE The 3 formulas above specify the computational steps required to implement a solution to the problem -
 It can be observed that:

- 1) many computations can be saved by properly storing intermediate results
- 2) dedicated matrix/vector operations (eg. in cross products) can save operations
- 3) The model does not ~~account~~ consider the data acquisition (image acquisition) or auxiliary processing - Since these steps could require significant time the model could require "corrections" (*) to ensure "accuracy" -
- 4) Suitable "sw engineering" is required to ensure
 - sw operating in real-time
 - suitable sw design
 - readability / maintainability
 - ...

(*) Q: which corrections?